Lattice architecture review

Platform team, Q1 2025

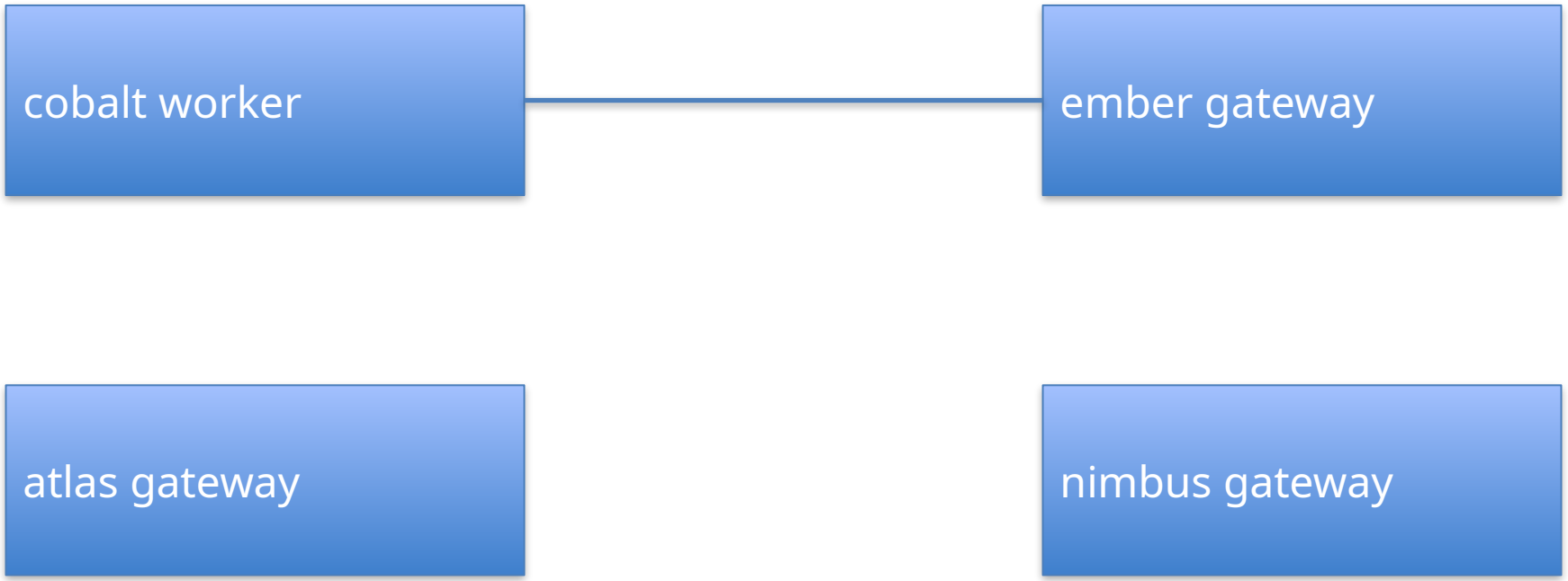
Failure handling (ember)

- Replicates hourly rollups so a failed region drains within minutes
- Signs outbound webhooks after the leader confirms the lease

Limits (saffron)

Service	p50 ms	p99 ms	Error rate
harbor-0	81	496	0.32%
juniper-1	69	344	0.33%
cobalt-2	56	1256	0.95%

Proposed design (saffron)



Cutover (atlas)

- Replicates pending writes after the leader confirms the lease
- Throttles audit records while the previous build still serves reads
- Archives tenant quotas before the nightly window closes
- Rejects incoming events while the previous build still serves reads